



Machine learning-aided prediction of nitrogen heterocycles in bio-oil from the pyrolysis of biomass

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ABSTRACT

Nitrogen heterocyclic compounds in bio-oil (NH₂Oil) made from biomass pyrolysis such as pyrroles, pyrazines, and indoles, have a relative content of 0–30%. NH₂Oil is a NO_x precursor if bio-oil is used as a fuel, but it has a high potential as a precursor for high-value chemicals. However, predicting and controlling NH₂Oil are challenging because of the complexity of the pyrolysis reaction system. Machine learning (ML) shows significant potential for addressing this issue. In this study, the relative contents of NH₂Oil, 5-membered NH₂Oil, 6-membered NH₂Oil, bio-oil yield, and the content of nitrogen in bio-oil were predicted using Random Forest and gradient boosting regression algorithms, with test regression coefficients values of 0.77–0.87 and 0.74–0.81 obtained for the former and latter ML models, respectively. Biomass N was the most important factor in predicting the bio-oil yield, whereas biomass N/C was the most significant of the other four targets and can be used as a proxy to assess the potential of biomass feedstock as fuel material or N-containing value-added chemical precursor. The optimization of pyrolysis parameters within ML models provides useful information for instructing experimental studies, indicating ML-aided bio-oil prediction and engineering show great promise and are worthy of further investigation.

1. Introduction

Nitrogen (N)-containing biomasses are a group of biomass feedstocks and biowastes, including algae, manure, food waste, and sewage sludge. In recent years, N-containing biomasses have attracted increasing attention from academia, governments, and enterprises because of the urgent and increasing demand for their disposal and valorization [1,2]. Billions of these biomasses are produced globally each year, posing a significant threat to the environment. Many technologies have been investigated for the disposal and valorization of these biomasses, including pyrolysis [3–5], hydrothermal carbonization [6–9], hydrothermal liquefaction [10], gasification [11], and torrefaction [12]. Pyrolysis has received great attention owing to its ability to process biomass in an oxygen-free/-deficient atmosphere at 300–900 °C to produce char (slow pyrolysis), oil (fast pyrolysis), or gas (high-temperature pyrolysis) products [13,14]. Oil products from pyrolysis, so-called bio-crude oil or bio-oil [15], have a leading edge in the field of biomass valorization research [16] because they can be further

processed to produce high-value products such as biofuels, biochemicals, and biomaterials [17,18].

The composition and characteristics of bio-oils determine their potential for use in the production of value-added biofuels, biochemicals, and biomaterials [19]. Among these components, N-containing chemicals in bio-oil are key to its utilization because these chemicals are the major components of bio-oil obtained from the pyrolysis of N-rich biomasses [20]. As mentioned in our previous review, the pyrolysis of N-rich biomasses produces a bio-oil with a high N content, up to 10% of its own weight [21]. The concentrations of major nitrogenous compounds in bio-oils (amines/amides, nitriles, and N-containing heterocycles) vary significantly depending on their processing conditions [21]. The importance of N-containing heterocyclic compounds or heterocycles (NH₂Oil) has already been established [22]. These compounds are further divided into 5-membered NH₂Oil (NH-5_Oil, including pyrroles and pyrrolines), 6-membered NH₂Oil (NH-6_Oil, including pyrazines and piperidines), and other NH₂Oil components (including indoles and aziridines). Various reaction pathways may contribute to the production

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of these N-containing compounds, including Maillard reactions, direct cracking of amino acids with heterocyclic structures, cyclization of long-chain aliphatic amino acids, cyclization/dimerization of amines/amides, dehydration of piperazines (e.g., DKPs), and secondary cracking of N-containing functional groups in char [21]. Both the composition of the biomass (e.g., protein and carbohydrate content) and the processing parameters (e.g., temperature) have considerable effects on NH₂Oil formation [21]. Therefore, it is difficult to determine the exact formation mechanisms of NH₂Oil, which increases the difficulty of predicting and mediating the NH₂Oil formation behavior.

NH₂Oil has two main applications. First, NH₂Oils are important NO_x precursors in the bio-oil combustion process [17,23,24]. Several studies have been conducted on denitrogenated bio-oil, where NH₂Oil was found to be stable but catalyst-poisoned, making the removal of the NH₂Oil-form N difficult [25,26]. Next, NH₂Oil shares a similar structure with other high-value chemicals, such as pharmaceutical precursors, hydrogen storage chemicals, and anti-corrosion agents. Therefore, it has a high potential to be used as a precursor of these chemicals [22,27,28]. For instance, 59% of pharmaceuticals approved by the United States Food and Drug Administration are nitrogen heterocycles. In particular, 5-/6-membered nitrogen heterocyclic species are dominant [29]. Most nitrogen heterocyclic species found in bio-oils have the potential to be used as liquid organic hydrogen carriers for on- or off-board hydrogen storage [30]. In addition, nitrogen heterocycles have been widely employed for corrosion mitigation in the aqueous phase because they can adsorb onto the metal surface and form a protective film that resists corrosion [31]. However, it is challenging to mediate NH₂Oil according to the various application goals of the as-produced bio-oil because of the complexity of the pyrolysis reaction system, as discussed earlier.

Machine learning (ML), one of the most widely used artificial intelligence (AI) technologies, is popular for addressing complex issues [26,32]. Recently, ML has been widely adopted for predicting the yields [33] and properties of products from thermochemical treatments of biomass, such as pyrolysis, hydrothermal liquefaction (HTL), and gasification [34–36]. For instance, the yields of bio-oil, gas, and char [37,38]; yield and H content of bio-oil [39]; higher heating value [40], specific surface area of biochar, and total pore volume of biochar from pyrolysis of biomass [28,41] have all been predicted by ML models at high accuracy (test regression coefficients (R^2) of 0.85–0.95). In our previous study, the Random Forest (RF) model was used to predict the relative content of nitrogen heterocycles in bio-oil from HTL biomass [22]. However, the predictive efficiency for nitrogen heterocycles was not favorable, with an R^2 value below 0.70. Even after integrating the bio-oil yield and N content of bio-oil as input variables, the R^2 value only improved slightly to 0.80, which was insufficient [22].

Many factors affect biomass pyrolysis products. However, experimental research is often limited to one or several types of raw materials/pyrolysis parameters, making it difficult to quantitatively evaluate the impact of pyrolysis on bio-oil. However, because of its strong data-fitting ability, ML is adept at solving complex nonlinear classification or regression problems in the field of biomass pyrolysis. Previously, researchers used ML to model and analyze biomass pyrolysis and achieved excellent results [42–44]. Therefore, ML can utilize experimental data for rapid and accurate predictions and optimizations, thereby saving time and cost and further contributing to the instruction of future experimental studies and bio-oil production.

To the best of our knowledge, NH₂Oil from biomass pyrolysis, which differs from NH₂Oil from HTL, has not yet been investigated. ML was used to predict the relative contents of NH₂Oil, NH-5_Oil, and NH-6_Oil as well as the bio-oil yield and N content in the bio-oil. This study aims to build accurate ML models that can be used to understand the NH₂Oil formation behavior and mechanisms (through model interpretations). The models are also expected to guide the experimental design and engineering of bio-oil chemical compositions (through model optimizations).

2. Methodologies

2.1. Data collection and correlation analysis

A total of 217 data points were collected from 63 SCI journal research articles comprising 91 biomasses, including algae, sewage sludge, animal excrement, and forestry or agricultural waste. Detailed information is provided in Tables S1 and S2 of the Supplementary Material. The input parameters for the biomass feedstocks included their elemental composition (C, H, O, N, and S), atomic mole ratio of elements (H/C, O/C, and N/C), and proximate analysis (volatile matter (VM), fixed carbon (FC), and ash). All the collected raw material data were considered on a dry basis (C + H + O + N + S + ash = 100). The pyrolysis conditions included the pyrolysis temperature and type of pyrolysis reaction (i.e., fast or slow pyrolysis, valued as 1 or 0, respectively, because the exact heating rate was not reported in previous studies). Thirteen variables were used as input variables. Meanwhile, five variables were targeted as outputs: the yield of pyrolyzed bio-oil calculated based on the yield of liquid products (Yield_Oil), nitrogen content of pyrolyzed bio-oil (N_Oil), N-containing heterocyclic compounds or heterocycles (NH₂Oil) content of pyrolyzed bio-oil, 5-membered NH₂Oil (NH-5_Oil) content of pyrolyzed bio-oil, and 6-membered NH₂Oil (NH-6_Oil) content of pyrolyzed bio-oil. The data for NH₂Oil, NH-5_Oil, and NH-6_Oil were derived from the gas chromatography-mass spectrometry (GC/MS) analyses of the pyrolyzed bio-oils, which were calculated according to the relative peak areas of all detected compounds, similar to previous ML studies [45]. When establishing the ML models, only complete datasets were used, implying that there were no missing values in the five sub-datasets for single-target prediction and one sub-dataset for multi-task prediction. All input data were available for the whole dataset, and the number of data in each sub-dataset (from approximately 100 to approximately 200) for predicting different targets depended on the availability of the five outputs (details provided in Supplementary Material), which are similar to those in previous ML studies in this domain [22,46–49]. A detailed distribution of the data and numerical ranges of the five targets under slow and fast pyrolysis is provided in the Supplementary Material (Figs. S1 and S2, respectively).

The correlation coefficient value between the input variables and the output variables was calculated using the Pearson correlation coefficient (PCC) [39,50], which was recorded as “ r ”. The value of r ranged between -1 and 1 . An r value greater than 0 was regarded as a positive correlation, whereas an r value less than 0 was regarded as a negative correlation. The r value was calculated using Eq. (1):

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (1)$$

where X_i and Y_i represent the data values collected, $\bar{X}$ and $\bar{Y}$ represent the average of these values, respectively.

In this work, the degree of correlation was set as follows: 0.00–0.10, negligible correlation; 0.10–0.39, weak correlation; 0.40–0.69, moderate correlation; and 0.70–0.89, strong correlation [51]. In addition, the significance of the correlation between each pair of input and output variables was tested. The PCC was used to detect the collinear relationship between any two input variables and the linear correlation between the input and output variables [41].

2.2. ML modeling and hyperparameters optimization

All ML algorithms and functions used in this study are based on various open-source libraries in Python 3.10.8. Specific ML algorithms include Random Forest (RF) and Gradient Boosting Regressors (GBR). As an integration algorithm, the RF contains multiple decision trees that can be used to study the nonlinear relationship between variables in

regression or classification problems [52,53]. RF has a relatively low requirement for data distribution, implying that it can handle data from various distribution states [54,55]. In addition, RF has several advantages, such as high accuracy, enhanced visualization [56], and its ability to smoothly process many input variables [36,57]. Likewise, GBR can be used for regression and classification research [58], which employs the gradient descent method to reduce the loss function and achieve a high accuracy [59]. GBR learns by integrating poorly-integrated learning algorithms, thereby converting weak learners into strong ones [60]. Compared with RF, GBR has a higher sensitivity to data distribution. The lower the training data fraction, the faster the regression, because the model is more suited for smaller data in each iteration [61].

The dataset was divided into training and testing datasets for each model; 80% of the dataset was used as the training dataset, whereas 20% was used as the test dataset. Before establishing the model, the data were normalized to limit the data within a certain range (namely between −1 and 1) to eliminate the negative impact caused by the large variation scales of different variables [62]. Subsequently, the hyperparameters “n_estimators” and “max_depth” [38], which correspond to the number of regression trees and the depth of trees, respectively, are crucial for establishing the prediction models [39]. In this study, “n_estimators” was tuned from 1 to 130, and “max_depth” from 1 to 20. The other hyperparameters remained unchanged. Cross-validation of the training dataset is a common strategy to improve computational efficiency and model accuracy, and to avoid overfitting [26]. In this study, the 5-fold cross-validation method was used during model tuning to determine the optimal hyperparameters. The training dataset was randomly divided into five parts, four of which were used in the model training to determine the optimal hyperparameters, whereas the remaining part was used to evaluate the performance. This process was repeated five times to ensure that all five parts were used to evaluate the performance, allowing the full utilization of the dataset for model training and testing. The final accuracy was obtained by averaging the accuracies across five processes [22]. After hyperparameter optimization, 11 models were established, including ten models (five RF models and five GBR models) for the single-task prediction of the five targets, namely Yield_Oil, N_Oil, NH_Oil, NH-5_Oil, and NH-6_Oil, and one RF model for the multi-task prediction of NH_Oil, NH-5_Oil, and NH-6_Oil.

2.3. ML model evaluation and interpretation

2.3.1. ML model evaluation

To evaluate the effectiveness of the ML prediction models, two validation factors, root mean square error (RMSE) and test regression coefficient (R^2) were used as the basic criteria to display the linearity between the predicted and actual data.

The RMSE represents the square root of the ratio of the square of the deviation between the predicted value and the real value and the number of observations. The RMSE was calculated using Eq. (2):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (2)$$

where y_i denotes the value predicted by the model, and $\hat{y}_i$ denotes the data value collected.

Meanwhile, the value of R^2 ranged between 0 and 1; an increase in this value corresponds to an increase in the degree of fit accuracy. The value of R^2 was calculated using Eq. (3):

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2} \quad (3)$$

where $\hat{y}_i$ represents the collected data, y_i is the value predicted by the model, and $\bar{y}$ is the average predicted value.

Residues analysis was conducted on all ML models, and the Kolmogorov-Smirnov (K-S) test (confidence interval = 95%) was

conducted for the obtained residues to calculate p -K-S [63,64]. If p -K-S > 0.05, the residues significantly obey the normal distribution, which provides positive evidence for the prediction feasibility of the ML model [65]. In addition, the probability density distribution of the relative errors (e) under different models was calculated using Eq. (4) to study the impact of the data distribution on the prediction performance.

$$e = \frac{Value_{exact} - Value_{predict}}{Value_{exact}} \quad (4)$$

2.3.2. ML model interpretation

The hyperparameters of the models were adjusted such that RMSE was minimized and R^2 was maximized [48]. A partial dependency graph (PDP) and Shapley additive explanations (SHAP) were used to explain the predictive models and analyze the relationship between the input and output variables. The PDP mainly displayed the important rankings and conducted a one-way partial correlation analysis to show the influence of important top-ranked input variables on the output. Further, the PDP can exhibit a functional relationship between one or two characteristic variables and the predicted results of the model, such as approximate linear relationships, monotonic relationships, or more complex relationships [57]. In addition, SHAP explains the output of ML models by translating the optimal credit distribution with value-based (local) explanations [66]. The concept of the Shapley value is based on game theory and can be used to evaluate the impact of input variables on the predicted output variables [22].

2.4. Model implementation for experimental studies

The multi-task RF model that simultaneously predicted the three variables (NH_Oil, NH-5_Oil, and NH-6_Oil) was then applied to examine the same variables under fast pyrolysis conditions, such as temperature variation. Six types of typical biomass feedstock collected from the literature were considered raw material data. The elemental composition, the atomic mole ratio of elements, proximate analysis, and temperature were used as input variables. Under fast pyrolysis, the pyrolysis temperature was increased from 300 °C to 900 °C with a step size of 10 °C. The above 6 biomasses yielded a large number of outputs (NH_Oil, NH-5_Oil, and NH-6_Oil), from which optimum solutions were obtained to help design experimental studies for bio-oil engineering [26].

3. Results and discussions

3.1. Data analysis

Fig. 1 shows the distribution of the collected data containing 20–60% C, 16–54% O, 0.3–12% N, 2.9–8.2% H, and 0–1.6% S, which represent the elemental contents of typical biomass types, particularly N-rich biomass [21,27] (Fig. 1a). The atomic ratios H/C (1.2–2.3) and O/C (0.3–1.2) (Fig. 1b) of almost all biomasses were higher than coal, with that of some biomasses comparable to lignite and peat, indicating the low aromaticity of biomass feedstocks [21,67]. Lower H/C and O/C values of biomass generally indicate a higher lignin content, which is the biochemical component with the highest aromaticity [68]. The N/C values ranged from zero to 0.199 because of the different N contents of the biomass, indicating the high efficiency of producing N-containing compounds. Similarly, the yield of VM, FC, and ash were 50–90%, 1–20%, and 0.3–40% (Fig. 1c), respectively, which also covered the proximate compositions of most biomass types [21]. A high VM yield is indicative of high cellulose and hemicellulose contents, whereas a high FC yield exhibits a high lignin content [68,69]. The values of temperature were collected from 300 to 700 °C (Fig. 1d), which is the typical temperature range for bio-oil production. Furthermore, the bio-oil yield ranged from 12% to 68% (Fig. 1e) because both the slow and fast pyrolysis datasets were used as input values. The N content in bio-oil ranged from 0.1 to 14.2%, as it was linearly related to the content of

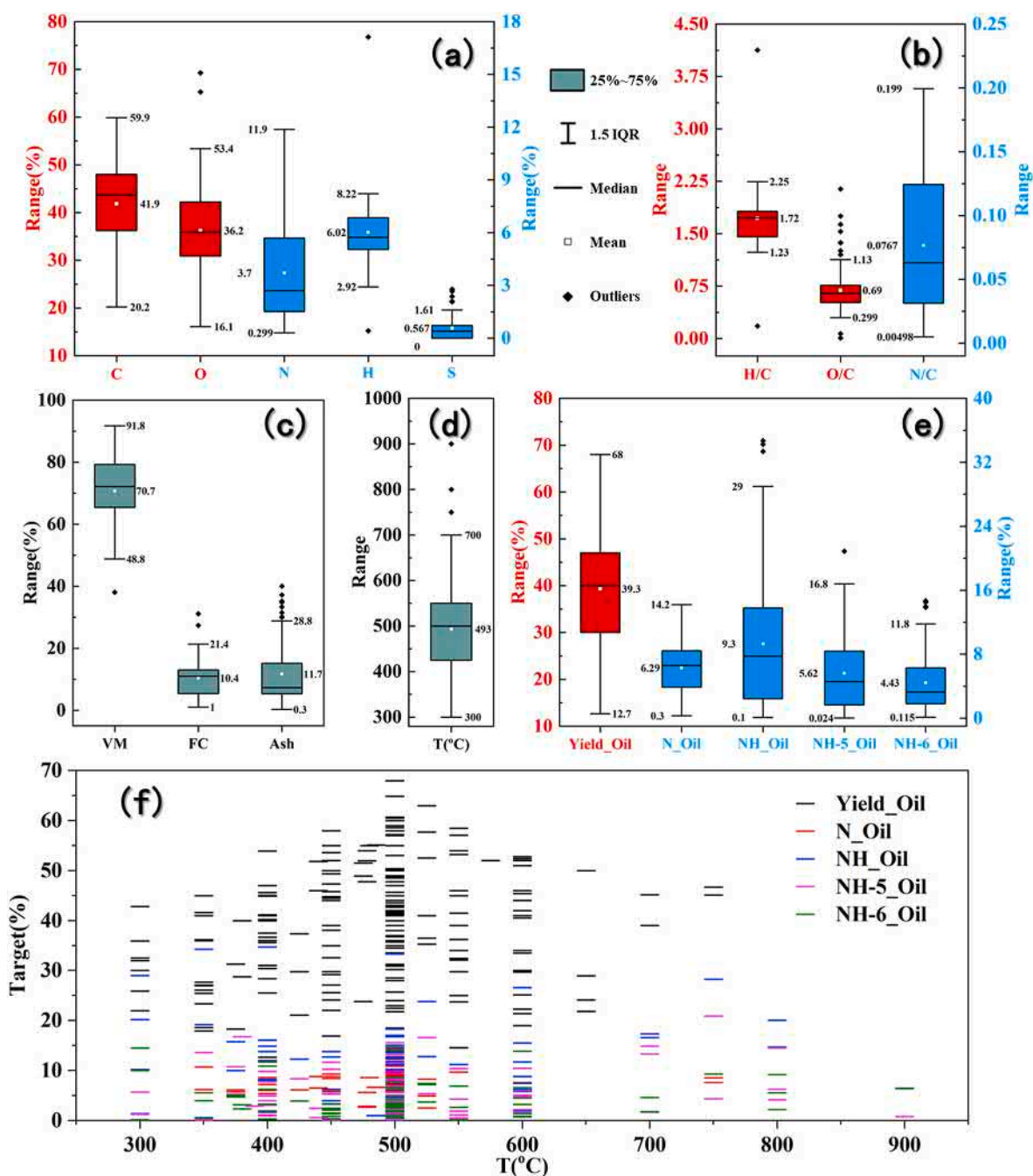


Fig. 1. Distributions of data for ML models: (a) elemental compositions, (b) atomic ratios, (c) proximate compositions, and (d–f) temperature and outputs (N_Oil, NH_Oil, NH-5_Oil, and NH-6_Oil) of the original data.

N in biomass; the linear correlation “bio-oil N = 1.14 * biomass N ($R^2 = 0.56$)” was obtained from a previous study [21], which determined that the N content of bio-oils was higher than the N content in biomass. NH_Oil constitutes 0.1–30% of the chemical compounds of bio-oil, with NH-5_Oil comprising approximately half of that, and NH-6_Oil comprising the remaining; the former was slightly higher than the latter (Fig. 1e), while other NH_Oil compounds such as indoles, quinolines, aziridines, and azetidines were detected at negligible amounts. Fig. 1f shows that the effects of temperature on the five targets were complex, and no clear trend could be identified, indicating that the biomass composition and other pyrolysis parameters should also have a considerable effect on these targets. Interestingly, the highest bio-oil yield was obtained between 400 and 600 °C. Interestingly, this temperature range did not necessarily result in a high NH_Oil value.

Generally, most of the data in this study were close to the normal distribution (Fig. S1).

According to the PCC (Fig. 2), NH_Oil only showed significant correlations with biomass N and N/C. NH-5_Oil showed similar correlations with these two parameters, but NH-6_Oil showed no significant correlations with these parameters. Notably, NH-6_Oil had a weak correlation with the reaction type, which resulted in a similar correlation between NH_Oil and the reaction type. Interestingly, NH-5_Oil exhibited a significant negative relationship with FC but a positive relationship with H/C (Fig. 2). In addition, N_Oil had the highest correlation values with biomass N and N/C because all N-containing bio-oil components were derived from biomass N. However, N_Oil showed a significant negative correlation with biomass O (Fig. 2), indicating that biomass N and O may compete for distribution into the bio-oil during pyrolysis.

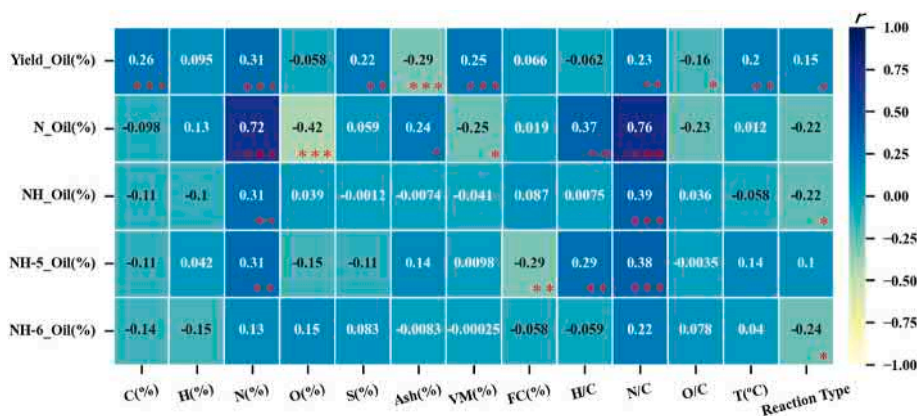


Fig. 2. PCC between the input and outputs (Number in each unit represents the r -value and asterisks denote the significance of the relationship; r : 0.00–0.10, negligible correlation; 0.10–0.39, weak correlation; 0.40–0.69, moderate correlation; 0.70–0.89, strong correlation; ***: $p < 0.001$; **: $0.001 < p < 0.01$; *: $0.01 < p < 0.05$).

Interestingly, many input variables were significantly related to Yield_Oil, including C, N, S, ash, VM, N/C, O/C, T, and the reaction type, implying that the prediction of Yield_Oil may be more complex than that of other targets (Fig. 2). Notably, the reaction type had only two values, namely, 0 (slow pyrolysis) and 1 (fast pyrolysis), and the latter produced more bio-oil (with a relatively low significance level, $r = 0.15$, $0.01 < p < 0.05$). According to the data variation of the five output variables under slow or fast pyrolysis, the different pyrolysis types (reaction types) showed a weak correlation with the Yield_Oil output. Moreover, the numerical ranges of the five predicted targets for the two types of pyrolysis were similar (Fig. S2). According to previous research, the bio-oil yield of fast pyrolysis is generally much higher than that of slow pyrolysis [70–72], with the former reaching up to 80% more [71,72]. Unfortunately, owing to the probable limitations of the dataset collected in this study, the impact of the pyrolysis type on the bio-oil cannot be determined conclusively. Another reason is that different biomass species were used for fast and slow pyrolysis, diminishing the comparison [73].

3.2. Single-task ML prediction and interpretation

Hyperparameters for all models were tuned and optimized with 5-cross validation, and the average validation RMSE (calculated based on normalized data) was used to screen the optimal hyperparameters; hyperparameters with the lowest RMSE were treated as the optimum. The models with optimal hyperparameters were trained and tested again using the training and testing datasets, respectively. Fig. 3 shows the optimal hyperparameters, predictive performances of the final RF ML models, and residues analyses or predictions for all five single-task RF prediction models. The blue area in Fig. 3 represents the confidence interval range of the training set in the RF model, whereas the red area represents the confidence interval range of the test set (confidence level = 95%). The residues analysis of the predicted values shows a distribution of residues near zero for the data points and confidence intervals. According to the K–S test results, the residues of the training and test sets for all RF models obeyed the normal distribution (p -K–S > 0.05, Fig. 3). Relevant information on the GBR models (Fig. S3) and the probability density distribution of the relative errors under different RF models is shown in the Supplementary Material (Fig. S4).

(1) Yield_Oil

As shown in Table 1 and Fig. 3a, the Yield_Oil was predicted using biomass and pyrolysis processes with test R^2 values of 0.80 and 0.74, for the RF (Fig. 3(a2)) and GBR (Fig. S3(a2)) models, respectively. Interestingly, RF outperformed GBR for the other four single-task predictions.

The RMSE (calculated based on reverting normalized data) value for these two models was approximately 5, which is close to the experimental error (5%) for the bio-oil yield reported in several studies [74–77]. Therefore, the predictive performance in this study is acceptable and comparable to previous ML studies with regard to bio-oil yield prediction [78]. However, previous ML studies achieved better predictive performance, with a test R^2 of ~ 0.90 and an RMSE value of 1–4 [37, 47,66].

Model interpretation was performed based on RF models. Fig. 4(a2) shows the top factors influencing Yield_Oil, ranked in descending order of importance: N, T, ash, S, and FC; notably, the reaction type was the least important. This differs slightly from the SHAP analysis, with T being the fourth most important feature and the reaction type being the sixth (Fig. 4(a1)). This can be attributed to the different calculation mechanisms used by these two methods; for example, SHAP applies a cooperative game-theoretic strategy during interpretation (for details, see Section 2). Figs. 4(a3–6) show the PDP based on the feature analysis results of Fig. 4a. It was observed that N and S have a similar effect on Yield_Oil (Figs. 4(a3, 6)), in which a continuous increase is observed followed by a plateau, indicating that N and S participated in the formation of the bio-oil components. Further, the Yield_Oil increased with increasing T, before peaking at approximately 500 °C, but further increases in T resulted in a gradual decrease in Yield_Oil (Fig. 4(a4)). This trend is consistent with many previous studies. However, ash was negatively related to Yield_Oil, with a steep slope when the ash content was lower than 8%; whereas the slope became rather gentle at higher ash contents (Fig. 4(a5)). This explains why the Yield_Oil was calculated without drying, and almost no ash (inorganic) was yielded in the bio-oil. These findings are consistent with the results of the PCC analysis. In addition, the points with significant relative errors were generally located at the left edge of the data distribution (Fig. S4a).

(2) N_Oil

The predictive efficiency of the RF model for N_Oil was slightly better than that for Yield_Oil, with a test R^2 value of 0.86 and an RMSE of 1.15 (Fig. 3(b2)). No previous studies have predicted the N content of bio-oil from pyrolysis. However, the test R^2 values of 0.61–0.75 with an RMSE of approximately 1.0 were obtained for the ML prediction of N_Oil from the HTL of biomass [35], and improved predictive performance for N_Oil from HTL (test R^2 of 0.87–0.90 with an RMSE of 0.6–0.8) was obtained when only algae were considered during modeling [36]. In addition, the contents of O and H and the atomic ratios such as H/C and O/C of bio-oil from pyrolysis of biomass were investigated, and test R^2 values of 0.75–0.85 were obtained from most ML models [39,47,78], which is consistent with the N_Oil prediction performance obtained in this study.

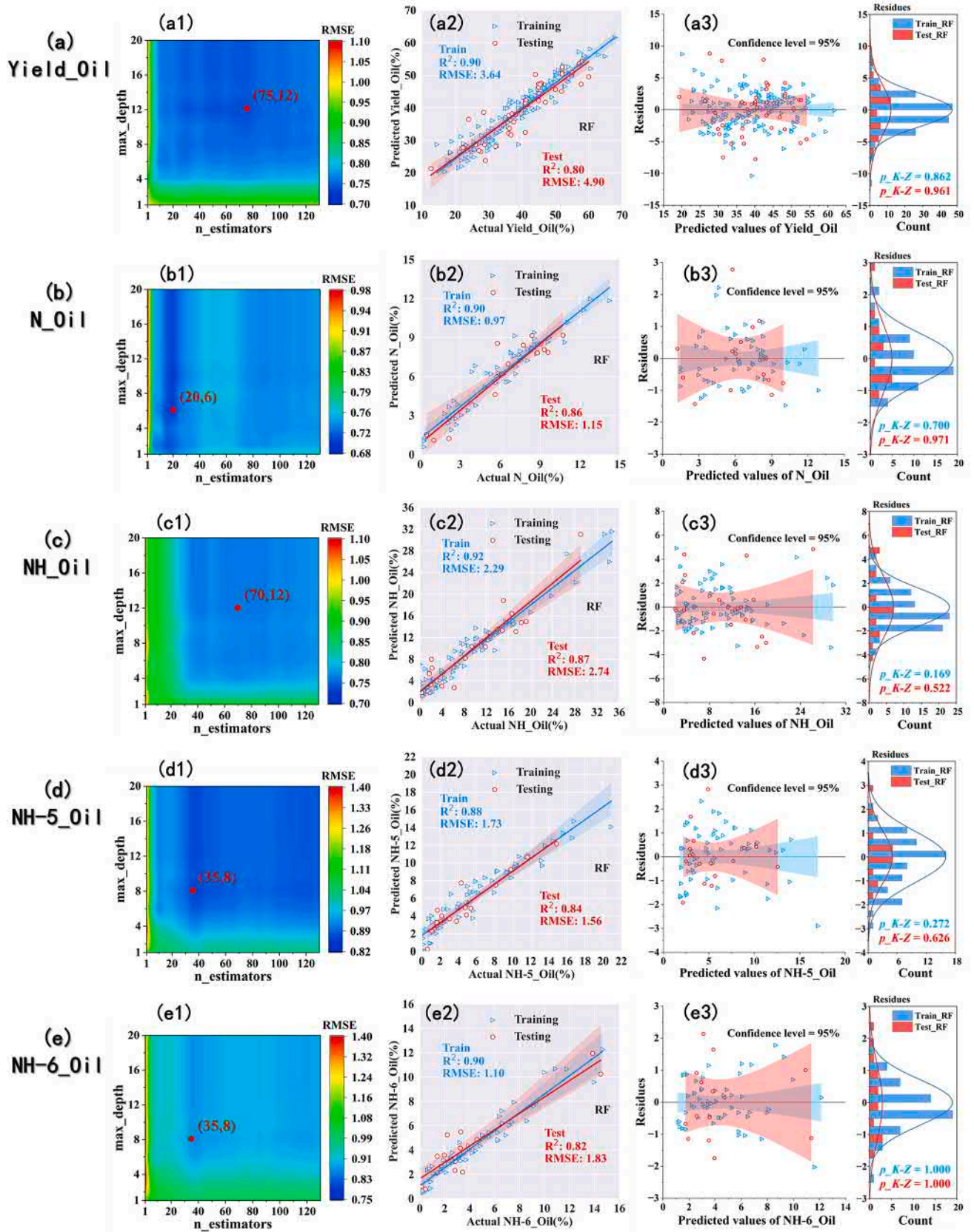


Fig. 3. Predictive performance of single-task RF models: (a) Yield_Oil, (b) N_Oil, (c) NH_Oil, (d) NH-5_Oil, (e) NH-6_Oil; (1) optimal hyperparameters, (2) result of model prediction, (3) residues analysis of predictions and K-S test on residues (p_K-Z , confidence level = 95%).

Table 1
Predictive performance of different ML tasks.

Model	Prediction targets ^a	Algorithms	Hyperparameters		Training		Testing	
			n_estimators	max_depth	R ²	RMSE	R ²	RMSE
1	Yield_Oil	GBR	30	4	0.93	3.11	0.74	5.56
2		RF	75	12	0.90	3.64	0.80	4.90
3	N_Oil	GBR	40	8	0.99	0.38	0.80	1.35
4		RF	20	6	0.90	0.97	0.86	1.15
5	NH_Oil	GBR	70	6	0.95	1.69	0.81	4.09
6		RF	70	12	0.92	2.29	0.87	2.74
7	NH-5_Oil	GBR	110	6	0.95	1.69	0.81	4.09
8		RF	35	8	0.88	1.73	0.84	1.56
9	NH-6_Oil	GBR	120	18	0.99	0.26	0.75	1.94
10		RF	35	8	0.90	1.10	0.82	1.83
11	NH_Oil, NH-5_Oil, NH-6_Oil	RF	75	14	0.88	1.70	0.77	2.17

^a Inputs used for prediction include elemental composition, atomic ratios of some elements, proximate analysis, and pyrolysis conditions (temperature and reaction type, slow or fast pyrolysis).

Both the important features and SHAP analysis predicted that N/C had a dominant effect on N_Oil (Figs. 4(b1-2)) and a gradual increase in N_Oil from 2.5% to 8.5% was observed when N/C increased from zero to 0.18 (Fig. 4(b3)). N/C can be a structural indicator, for example, H/C and O/C, which represent the aromaticity and stability of a material such as biochar [79,80], and the N/C of biomass determines the distribution of N to bio-oil because a higher N/C indicates a higher content of N in carbon structures for various reactions, such as the Maillard reaction, which covers a series of sub-reactions between the amino groups of amino acids and the O-containing functional groups of sugars [21,81]. Although the PCC indicated that N, O, and H/C were significantly correlated with N_Oil (Fig. 2), the other variables (input) were negligibly correlated. This can be explained by two reasons: 1) PCC analysis only illustrates linear relationships between two variables, whereas feature importance and SHAP analyses are tools to explain “black-box” ML models by identifying deep relationships between the inputs and outputs; 2) the importance levels of all features in the latter two analysis methods are complementary, unlike the independence characteristic of correlations in PCC. For example, N was a descriptor like N/C for biomass; however, N had only a supplementary effect on describing the potential of biomass for N_Oil, thus showing a considerably low importance level (Figs. 4(b1-2)), whereas, in the PCC, both N and N/C were independently related to N_Oil with a high significance level (Fig. 2).

(3) NH_Oil

The predictive efficiency for NH_Oil by RF, with a test R^2 value of 0.87 and RMSE of 2.74, was similar to that of N_Oil (Figs. 3(b2) and (c2)), and these two analysis methods also showed that N/C was the most important feature (Figs. 4(c1-2)), with an influential trend on NH_Oil (Fig. 4(c3)), similar to that on N_Oil. To date, the ML-prediction of NH_Oil in bio-oil from pyrolysis is yet to be reported, but a previous ML study dealing with HTL of biomass predicted NH_Oil with a test R^2 value of 0.59–0.67, and the incorporation of N_Oil as an input variable increased the test R^2 value to 0.75–0.82 [22]. The above-mentioned and present studies propose that NH_Oil is the major constituent of N_Oil in bio-oil, either from pyrolysis or HTL. Owing to the high predictive efficiency of this study, the incorporation of N_Oil as an input variable was unnecessary for predicting NH_Oil. In addition, other parameters such as ash, H/C, and T had negative effects on NH_Oil (Figs. 4(c4-6)).

(4) NH-5_Oil and NH-6_Oil

The predicted performances of NH-5_Oil (test $R^2 = 0.84$ for the RF model, Fig. 3(d2)) and NH-6_Oil (test $R^2 = 0.82$, Fig. 3(e2)), which were the dominant components of N_Oil and NH_Oil, were comparable to those of N_Oil and NH_Oil. The test RMSE values for these models were

1.5–2.0, indicating a low margin of error for the NH-5_Oil and NH-6_Oil predictions. N/C remained the most significant feature according to the SHAP analysis (Figs. 4(d1) and (e1)); however, the feature importance analysis showed that N/C ranked second behind H/C for predicting NH-5_Oil (Fig. 4(d2)) and third behind S and O for predicting NH-6_Oil (Fig. 4(e2)). This indicates that specific N-containing heterocyclic structures were formed in bio-oil, probably from different reactions, with considerable contributions from other structural characteristics of the biomass. High H/C and low FC (low lignin content of biomass [68]) favored the formation of NH-5_Oil (Figs. 4(d3,6)), implying that cellulose and hemicellulose enhanced the formation of NH-5_Oil, most probably through the Maillard reaction [21,81].

Interestingly, NH-5_Oil peaked at approximately 750 °C. This trend may appear erratic, as limited data have been reported in this range. Similar findings were also observed for NH-6_Oil, where the Maillard reaction is a major reaction that accounts for its formation, and low FC and high O (which determine the availability of O functional groups for the Maillard reaction) contributed considerably to NH-6_Oil formation (Figs. 4(e4,6)). The abnormal peak for the FC in Fig. 4(e6) may be attributed to the influence of specific cases, and this trend may not be authentic. The difference in the feature importance analysis between NH-5_Oil and NH-6_Oil indicated that the specific structures accounting for their formation during biomass pyrolysis varied considerably, even though both products were formed through Maillard reaction-related paths.

The results of the residues analysis indicated that the predictions of both models were reliable (Figs. 3(d3) and (e3)). Moreover, the analysis results for the relative errors of these two models are similar to those of the other RF models, showing large relative errors at the left edge of the data distribution (Figs. S4(b–e)), which is consistent with results obtained from previous studies [82]. This can be attributed to the small data values at the left edge of the distribution and the small amount of data [81].

3.3. Multi-task ML model prediction and application

(1) Multi-task ML model predictions

The multi-task prediction of multiple targets (NH_Oil, NH-5_Oil, and NH-6_Oil) was achieved with a test R^2 value of 0.77 and a test RMSE value of 2.17 (Fig. 5c). The residues analysis was represented by the total residues values of the three target variables, and both the residues of the training and test sets for this model significantly obeyed the normal distribution (p -K-S > 0.05, Fig. 5b). In descending order, the top features for the three targets of importance are FC, T, N/C, and H/C in the feature analysis (Fig. 5e) and H/C, N/C, FC, and T in the SHAP analysis (Fig. 5d). These lists differed slightly from those in previous single-task prediction models because this multi-task model balances trade-offs by

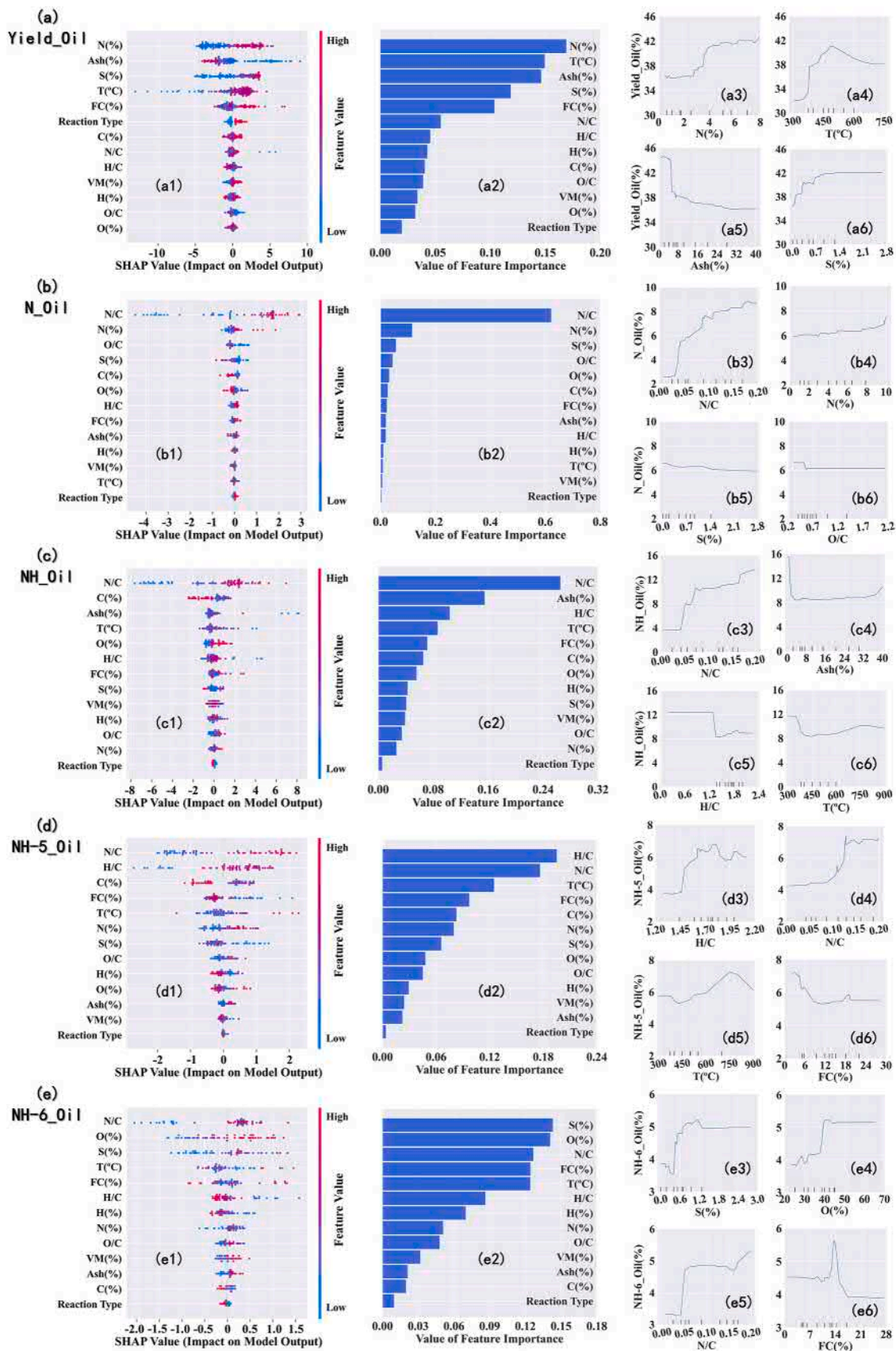


Fig. 4. Interpretation of the single-task RF models: (a) Yield_Oil, (b) N_Oil, (c) NH_Oil, (d) NH-5_Oil, (e) NH-6_Oil; (1–2) the sequence of feature importance in RF model analyzed by SHAP and PDP, (3–6) impact of the four most significant features in the RF model analyzed by PDP.

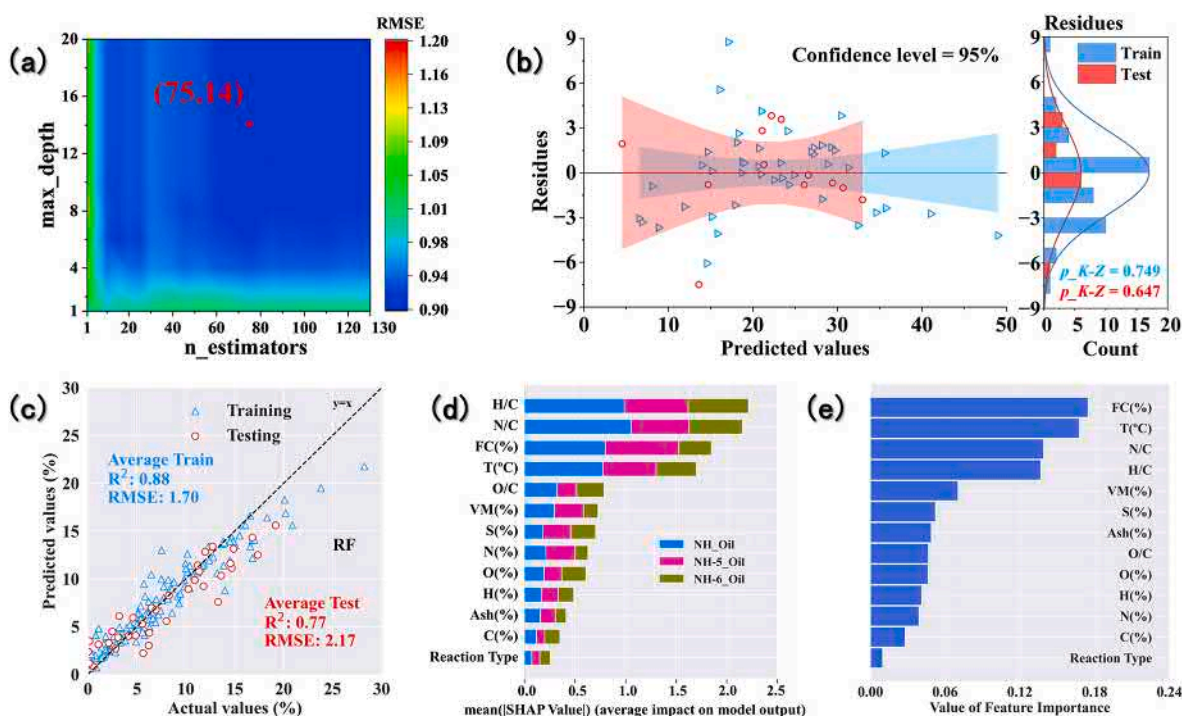


Fig. 5. Predictive performance and the interpretation of the multi-task RF model for predicting NH_Oil, NH-5_Oil, and NH-6_Oil: (a) optimal hyperparameters, (b) residues analysis of predictions and K-S test on residues (p_{K-Z} , confidence level = 95%), (c) the result of RF model prediction, (d-e) the sequence of feature importance analyzed by SHAP or PDP.

tuning the hyperparameters among the three targets, and equal weights were used during modeling. Considering that the amount of valid data in the dataset in multi-task predictions was significantly smaller than that in single-task predictions, the model interpretations for every single target from single-task prediction models were more reliable.

(2) Model implementation for the experimental studies

A multi-task model was used to optimize the NH_Oil composition in the bio-oil. Six different biomasses were used to evaluate the optimal pyrolysis conditions by incorporating the compositions of these feedstocks at variable pyrolysis temperatures ranging from 300 to 900 °C with a step size of 10 °C. Detailed information is provided in Table S3 of the Supplementary Material. The optimization results are shown in Fig. 6. The biomass type had the most significant impact on NH-5_Oil and NH-6_Oil; for example, Raspberry Seed Cake yielded the lowest

NH_Oil, Manure yielded the highest NH-6_Oil, while Rapeseed Oil Cake yielded the highest NH-5_Oil. While the temperature considerably influenced the composition of NH-5_Oil and NH-6_Oil, its influence was not as significant as that of the biomass type. For example, for the sewage sludge and *Chlorella Vulgaris*, higher temperatures led to lower NH-5_Oil and NH-6_Oil outputs, whereas lower temperatures led to higher NH-5_Oil, NH-6_Oil, and NH_Oil outputs (Fig. 6).

The prediction results and optimal hyperparameters obtained from the ML model provided valuable practical guidance. The final ML model was built for practical applications, and pre-testing was useful for verifying the model. If the ML model performs poorly in practice, it should be retrained using the experimental data. However, the accuracy of this multi-task ML model was not very high ($R^2 < 0.8$), and the dataset size for constructing this ML model was small, indicating the low generalizability of the model. Thus, experimental verification was not conducted, and a more generalizable and powerful model is needed for a

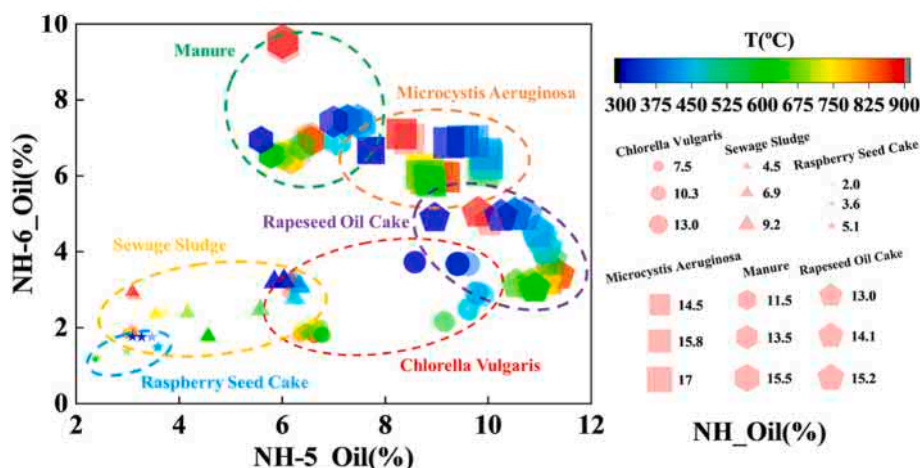


Fig. 6. Results of NH_oil composition optimization in bio-oil.

better understanding of the experimental results. Therefore, this study indicates that ML-aided bio-oil prediction and engineering are promising alternatives to time-consuming traditional methodologies, such as the trial-and-error method, it is thereby worthy of further investigation.

4. Conclusions

The data distribution of the dataset used in this study was similar to the normal distribution; data points with large relative errors are usually located at the boundary of the data distribution, and the prediction effect in the central region with a more concentrated data distribution is more accurate. The single-task ML accurately predicted the relative contents of NH₄Oil, NH₄-5Oil, and NH₄-6, as well as Yield_{Oil} and N_{Oil} (with a test R^2 of 0.80–0.87 and an RMSE of 1.00–5.00) when using biomass compositions and pyrolysis parameters as inputs. N was the most significant input for predicting Yield_{Oil} because it promotes the formation of bio-oil chemicals. In addition, the N/C ratio was the most significant input affecting the other four outputs and can be used as a proxy to determine the potential of biomass feedstock as a fuel material or N-value-added chemical precursor. Finally, the multi-task ML model (with an acceptable R^2 of 0.77) developed in this study successfully predicted NH₄Oil, NH₄-5Oil, and NH₄-6 simultaneously, and can guide future experimental studies in related areas.

However, this study has some limitations. First, the biochemical compositions of the biomass (lignin, cellulose, hemicellulose, lipid, and protein) and pyrolysis parameters (i.e., heating rate, purge gas type, flow rate, and reactor characteristics) were not considered because of the insufficient availability of these descriptors or fingerprints in previous studies. Moreover, only the relative contents of NH₄Oil were available, rather than its quantitative composition, and the dataset size for ML modeling was limited. These were the main reasons for the imperfect predictive performance observed in this study. Future ML studies should aim to establish a dataset with better distribution, larger data volume, and wider data distribution to mitigate the adverse impact of data distribution on the ML model prediction performance and consider these to obtain improved predictive performance, a more in-depth understanding of NH₄Oil formation mechanisms, and better experimental instructions with verifications.

Credit author statement

Lijian Leng: Conceptualization, Supervision, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition; **Tanghao Li:** Investigation, Data curation, Writing – original draft, Writing – review & editing; **Hao Zhan:** Validation; **Muhammad Rizwan:** Writing – review & editing; **Weijin Zhang:** Writing – review & editing; **Haoyi Peng:** Resources; **Zequn Yang:** Formal analysis, Writing – review & editing; **Hailong Li:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2023.127967>.

References

- [1] Skaggs RL, Coleman AM, Seiple TE, Milbrandt AR. Waste-to-Energy biofuel production potential for selected feedstocks in the conterminous United States. *Renew Sustain Energy Rev* 2018;82:2640–51. <https://doi.org/10.1016/j.rser.2017.09.107>.
- [2] Leng L, Li J, Yuan X, Li J, Han P, Hong Y, et al. Beneficial synergistic effect on bio-oil production from co-liquefaction of sewage sludge and lignocellulosic biomass. *Bioreour Technol* 2018;251:49–56. <https://doi.org/10.1016/j.biortech.2017.12.018>.
- [3] Wang S, Dai G, Yang H, Luo Z. Lignocellulosic biomass pyrolysis mechanism: a state-of-the-art review. *Prog Energy Combust Sci* 2017;62:33–86. <https://doi.org/10.1016/j.peccs.2017.05.004>.
- [4] Chen D, Cen K, Zhuang X, Gan Z, Zhou J, Zhang Y, et al. Insight into biomass pyrolysis mechanism based on cellulose, hemicellulose, and lignin: evolution of volatiles and kinetics, elucidation of reaction pathways, and characterization of gas, biochar and bio-oil. *Combust Flame* 2022;242. <https://doi.org/10.1016/j.combustflame.2022.112142>.
- [5] Huang Y, Li B, Liu D, Xie X, Zhang H, Sun H, et al. Fundamental advances in biomass autothermal/oxidative pyrolysis: a review. *ACS Sustainable Chem Eng* 2020;8:11888–905. <https://doi.org/10.1021/acssuschemeng.0c04196>.
- [6] Wang T, Zhai Y, Zhu Y, Li C, Zeng G. A review of the hydrothermal carbonization of biomass waste for hydrochar formation: process conditions, fundamentals, and physicochemical properties. *Renew Sustain Energy Rev* 2018;90:223–47. <https://doi.org/10.1016/j.rser.2018.03.071>.
- [7] Xu ZX, Ma XQ, Zhou J, Duan PG, Zhou WY, Ahmad A, et al. The influence of key reactions during hydrothermal carbonization of sewage sludge on aqueous phase properties: a review. *J Anal Appl Pyrolysis* 2022;167. <https://doi.org/10.1016/j.jaap.2022.105678>.
- [8] Li B, Zhao L, Xie X, Lin D, Xu H, Wang S, et al. Volatile-char interactions during biomass pyrolysis: effect of char preparation temperature. *Energy* 2021;215. <https://doi.org/10.1016/j.energy.2020.119189>.
- [9] Li B, Tang J, Xie X, Wei J, Xu D, Shi L, et al. Char structure evolution during molten salt pyrolysis of biomass: effect of temperature. *Fuel* 2023;331. <https://doi.org/10.1016/j.fuel.2022.125747>.
- [10] Huang HJ, Yuan XZ. Recent progress in the direct liquefaction of typical biomass. *Prog Energy Combust Sci* 2015;49:59–80. <https://doi.org/10.1016/j.peccs.2015.01.003>.
- [11] Molino A, Chianese S, Musmarra D. Biomass gasification technology: the state of the art overview. *J Energy Chem* 2016;25:10–25. <https://doi.org/10.1016/j.jechem.2015.11.005>.
- [12] Dai L, Wang Y, Liu Y, Ruan R, He C, Yu Z, et al. Integrated process of lignocellulosic biomass torrefaction and pyrolysis for upgrading bio-oil production: a state-of-the-art review. *Renew Sustain Energy Rev* 2019;107:20–36. <https://doi.org/10.1016/j.rser.2019.02.015>.
- [13] Chen J, Wang P, Ding L, Yu T, Leng S, Chen J, et al. The comparison study of multiple biochar stability assessment methods. *J Anal Appl Pyrolysis* 2021;156. <https://doi.org/10.1016/j.jaap.2021.105070>.
- [14] Leng L, Yang L, Chen J, Hu Y, Li H, Li H, et al. Valorization of the aqueous phase produced from wet and dry thermochemical processing biomass: a review. *J Clean Prod* 2021;294. <https://doi.org/10.1016/j.jclepro.2021.126238>.
- [15] Chen D, Zhuang X, Gan Z, Cen K, Ba Y, Jia D. Co-pyrolysis of light bio-oil leached bamboo and heavy bio-oil: effects of mass ratio, pyrolysis temperature, and residence time on the biochar. *Chem Eng J* 2022;437. <https://doi.org/10.1016/j.cej.2022.135253>.
- [16] Li B, Song M, Xie X, Wei J, Xu D, Ding K, et al. Oxidative fast pyrolysis of biomass in a quartz tube fluidized bed reactor: effect of oxygen equivalence ratio. *Energy* 2023;270. <https://doi.org/10.1016/j.energy.2023.126987>.
- [17] Leng L, Li H, Yuan X, Zhou W, Huang H. Bio-oil upgrading by emulsification/microemulsification: a review. *Energy* 2018;161:214–32. <https://doi.org/10.1016/j.energy.2018.07.117>.
- [18] Yang L, Wang G, Liu T, Wan Y, Peng H, Leng L, et al. Experimental and ReaxFF molecular dynamic study on pyrolysis of phenylalanine. *Fuel* 2022;324. <https://doi.org/10.1016/j.fuel.2022.124690>.
- [19] Liu Z, Cui Y, Wang J, Yue C, Agbodjan YS, Yang Y. Multi-objective optimization of multi-energy complementary integrated energy systems considering load prediction and renewable energy production uncertainties. *Energy* 2022;254. <https://doi.org/10.1016/j.energy.2022.124399>.
- [20] Zhao C, Hong C, Hu J, Xing Y, Ling W, Zhang B, et al. Upgrading technologies and catalytic mechanisms for heteroatomic compounds from bio-oil – a review. *Fuel* 2023;333:126388. <https://doi.org/10.1016/j.fuel.2022.126388>.
- [21] Leng L, Yang L, Chen J, Leng S, Li H, Li H, et al. A review on pyrolysis of protein-rich biomass: nitrogen transformation. *Bioreour Technol* 2020;315. <https://doi.org/10.1016/j.biortech.2020.123801>.
- [22] Leng L, Zhang W, Chen Q, Zhou J, Peng H, Zhan H, et al. Machine learning prediction of nitrogen heterocycles in bio-oil produced from hydrothermal liquefaction of biomass. *Bioreour Technol* 2022;362. <https://doi.org/10.1016/j.biortech.2022.127791>.

- [23] Broumand M, Albert-Green S, Yun S, Hong Z, Thomson MJ. Spray combustion of fast pyrolysis bio-oils: applications, challenges, and potential solutions. *Prog Energy Combust Sci* 2020;79. <https://doi.org/10.1016/j.pecs.2020.100834>.
- [24] Chen B, Liu P, Pelucchi M, Guidici C, Maffei LP, Faller S, et al. New insights into the oxidation chemistry of pyrrole, an N-containing biomass tar component. *Proc Combust Inst* 2022. <https://doi.org/10.1016/j.proci.2022.07.019>.
- [25] Xu D, Lin G, Guo S, Wang S, Guo Y, Jing Z. Catalytic hydrothermal liquefaction of algae and upgrading of biocrude: a critical review. *Renew Sustain Energy Rev* 2018;97:103–18. <https://doi.org/10.1016/j.rser.2018.08.042>.
- [26] Li H, Chen J, Zhang W, Zhan H, He C, Yang Z, et al. Machine-learning-aided thermochemical treatment of biomass: a review. *Biofuel Research Journal* 2023;10: 1786–809. <https://doi.org/10.18331/brj2023.10.1.4>.
- [27] Leng L, Zhang W, Peng H, Li H, Jiang S, Huang H. Nitrogen in bio-oil produced from hydrothermal liquefaction of biomass: a review. *Chem Eng J* 2020;401. <https://doi.org/10.1016/j.cej.2020.126030>.
- [28] Leng L, Zhou J, Li T, Vlaskin M, Zhan H, Peng H, et al. Nitrogen heterocycles in bio-oil produced from hydrothermal liquefaction of biomass: a review. *Fuel* 2023;335: 126995. <https://doi.org/10.1016/j.fuel.2022.126995>.
- [29] Vitaku E, Smith DT, Njardarson JT. Analysis of the structural diversity, substitution patterns, and frequency of nitrogen heterocycles among U.S. FDA approved pharmaceuticals. *J Med Chem* 2014;57:10257–74. <https://doi.org/10.1021/jm501100b>.
- [30] Tan KC, He T, Chua YS, Chen P. Recent advances of catalysis in the hydrogenation and dehydrogenation of N-heterocycles for hydrogen storage. *J Phys Chem C* 2021; 125:18553–66. <https://doi.org/10.1021/acs.jpcc.1c04783>.
- [31] Verma C, Abdellatif MH, Alfantazi A, Quraishi MA. N-heterocycle compounds as aqueous phase corrosion inhibitors: a robust, effective and economic substitute. *J Mol Liq* 2021;340. <https://doi.org/10.1016/j.molliq.2021.117211>.
- [32] Lu J, Xiong R, Tian J, Wang C, Sun F. Deep learning to estimate lithium-ion battery state of health without additional degradation experiments n.d. <https://doi.org/10.1038/s41467-023-38458-w>.
- [33] Kovács DP, McCorkindale W, Lee AA. Quantitative interpretation explains machine learning models for chemical reaction prediction and uncovers bias. *Nat Commun* 2021;12. <https://doi.org/10.1038/s41467-021-12895-w>.
- [34] Ascher S, Watson I, You S. Machine learning methods for modelling the gasification and pyrolysis of biomass and waste. *Renew Sustain Energy Rev* 2022;155. <https://doi.org/10.1016/j.rser.2021.111902>.
- [35] Li J, Zhang W, Liu T, Yang L, Li H, Peng H, et al. Machine learning aided bio-oil production with high energy recovery and low nitrogen content from hydrothermal liquefaction of biomass with experiment verification. *Chem Eng J* 2021;425. <https://doi.org/10.1016/j.cej.2021.130649>.
- [36] Zhang W, Li J, Liu T, Leng S, Yang L, Peng H, et al. Machine learning prediction and optimization of bio-oil production from hydrothermal liquefaction of algae. *Bioresour Technol* 2021;342. <https://doi.org/10.1016/j.biortech.2021.126011>.
- [37] Leng E, He B, Chen J, Liao G, Ma Y, Zhang F, et al. Prediction of three-phase product distribution and bio-oil heating value of biomass fast pyrolysis based on machine learning. *Energy* 2021;236. <https://doi.org/10.1016/j.energy.2021.121401>.
- [38] Li H, Ai Z, Yang L, Zhang W, Yang Z, Peng H, et al. Machine learning assisted predicting and engineering specific surface area and total pore volume of biochar. *Bioresour Technol* 2023;369. <https://doi.org/10.1016/j.biortech.2022.128417>.
- [39] Tang Q, Chen Y, Yang H, Liu M, Xiao H, Wu Z, et al. Prediction of bio-oil yield and hydrogen contents based on machine learning method: effect of biomass compositions and pyrolysis conditions. *Energy Fuel* 2020;34:11050–60. <https://doi.org/10.1021/acs.energyfuels.0c01893>.
- [40] Chen J, Ding L, Wang P, Zhang W, Li J, Mohamed BA, et al. The estimation of the higher heating value of biochar by data-driven modeling. *J Renew Mater* 2022;10: 1555–74. <https://doi.org/10.32604/jrm.2022.018625>.
- [41] Leng L, Yang L, Lei X, Zhang W, Ai Z, Yang Z, et al. Machine learning predicting and engineering the yield, N content, and specific surface area of biochar derived from pyrolysis of biomass. *Biochar* 2022;4:63. <https://doi.org/10.1007/s42773-022-00183-w>.
- [42] Zhu X, Liu B, Sun L, Li R, Deng H, Zhu X, et al. Machine learning-assisted exploration for carbon neutrality potential of municipal sludge recycling via hydrothermal carbonization. *Bioresour Technol* 2023;369. <https://doi.org/10.1016/j.biortech.2022.128454>.
- [43] Haider Jaffari Z, Jeong H, Shin J, Kwak J, Son C, Lee Y-G, et al. Machine-learning-based prediction and optimization of emerging contaminants' adsorption capacity on biochar materials. *Chem Eng J* 2023;466:143073. <https://doi.org/10.1016/j.cej.2023.143073>.
- [44] Shafizadeh A, Shahbeik H, Rafiee S, Moradi A, Shahbaz M, Madadi M, et al. Machine learning-based characterization of hydrochar from biomass: implications for sustainable energy and material production. *Fuel* 2023;347. <https://doi.org/10.1016/j.fuel.2023.128467>.
- [45] Subramanya SM, Rios N, Kollar A, Stofanek R, Maloney K, Waltz K, et al. Statistical models for predicting oil composition from hydrothermal liquefaction of biomass. *Energy Fuel* 2023. <https://doi.org/10.1021/acs.energyfuels.3c00297>.
- [46] Cao H, Milan YJ, Mood SH, Ayiania M, Zhang S, Gong X, et al. A novel elemental composition based prediction model for biochar aromaticity derived from machine learning. *Artificial Intelligence in Agriculture* 2021;5:133–41. <https://doi.org/10.1016/j.aiia.2021.06.002>.
- [47] Zhang T, Cao D, Feng X, Zhu J, Lu X, Mu L, et al. Machine learning prediction of bio-oil characteristics quantitatively relating to biomass compositions and pyrolysis conditions. *Fuel* 2022;312. <https://doi.org/10.1016/j.fuel.2021.122812>.
- [48] Djandja OS, Duan PG, Yin LX, Wang ZC, Duo J. A novel machine learning-based approach for prediction of nitrogen content in hydrochar from hydrothermal carbonization of sewage sludge. *Energy* 2021;232. <https://doi.org/10.1016/j.energy.2021.121010>.
- [49] Djandja OS, Salami AA, Wang ZC, Duo J, Yin LX, Duan PG. Random forest-based modeling for insights on phosphorus content in hydrochar produced from hydrothermal carbonization of sewage sludge. *Energy* 2022;245. <https://doi.org/10.1016/j.energy.2022.123295>.
- [50] Baak M, Koopman R, Snoek H, Klous S. A new correlation coefficient between categorical, ordinal and interval variables with Pearson characteristics. *Comput Stat Data Anal* 2020;152. <https://doi.org/10.1016/j.csda.2020.107043>.
- [51] Schober P, Schwarte LA. Correlation coefficients: appropriate use and interpretation. *Anesth Analg* 2018;126:1763–8. <https://doi.org/10.1213/ANE.0000000000002864>.
- [52] Breiman L. Random forests. *Mach Learn* 2001;45:5–32. <https://doi.org/10.1023/A:1010933404324>.
- [53] Sun Y, Cheng H, Zhang S, Mohan MK, Ye G, De Schutter G. Prediction & optimization of alkali-activated concrete based on the random forest machine learning algorithm. *Construct Build Mater* 2023;385. <https://doi.org/10.1016/j.conbuildmat.2023.131519>.
- [54] Scornet E, Biau G, Vert JP. Consistency of random forests. *Ann Stat* 2015;43: 1716–41. <https://doi.org/10.1214/15-AOS1321>.
- [55] Gomes HM, Bifet A, Read J, Barddal JP, Enembreck F, Pfahringer B, et al. Adaptive random forests for evolving data stream classification. *Mach Learn* 2017;106: 1469–95. <https://doi.org/10.1007/s10994-017-5642-8>.
- [56] Hidayat F, Astsauri TMS. Applied random forest for parameter sensitivity of low salinity water Injection (LSWI) implementation on carbonate reservoir. *Alex Eng J* 2022;61:2408–17. <https://doi.org/10.1016/j.aej.2021.06.096>.
- [57] Ullah Z, Khan M, Raza Naqvi S, Farooq W, Yang H, Wang S, et al. A comparative study of machine learning methods for bio-oil yield prediction – a genetic algorithm-based features selection. *Bioresour Technol* 2021;335. <https://doi.org/10.1016/j.biortech.2021.125292>.
- [58] Natekin A, Knoll A. Gradient boosting machines, a tutorial. *Front Neurobot* 2013;7. <https://doi.org/10.3389/fnbot.2013.00021>.
- [59] Zhao W, Li J, Zhao J, Zhao D, Zhu X. PDD-GBR: research on evaporation duct height prediction based on gradient boosting regression algorithm. *Radio Sci* 2019; 54:949–62. <https://doi.org/10.1029/2019RS006882>.
- [60] Singh HM, Sharma DP, Alade IO. GBR-GSO based machine learning predictive model for estimating density of Al₂N₃, Si₃N₄, and TiN nanoparticles suspended in ethylene glycol nanofluids. *Eur Phys J Plus* 2022;137. <https://doi.org/10.1140/epjp/s13360-022-02767-8>.
- [61] Rathakrishnan V, Beddu S Bt, Ahmed AN. Predicting compressive strength of high-performance concrete with high volume ground granulated blast-furnace slag replacement using boosting machine learning algorithms. *Sci Rep* 2022;12. <https://doi.org/10.1038/s41598-022-12890-2>.
- [62] Zhang W, Chen Q, Chen J, Xu D, Zhan H, Peng H, et al. Machine learning for hydrothermal treatment of biomass: a review. *Bioresour Technol* 2023;370. <https://doi.org/10.1016/j.biortech.2022.128547>.
- [63] Sarmadi H, Karamodin A. A novel anomaly detection method based on adaptive Mahalanobis-squared distance and one-class KNN rule for structural health monitoring under environmental effects. *Mech Syst Signal Process* 2020;140. <https://doi.org/10.1016/j.ymssp.2019.106495>.
- [64] Virkar Y, Clauset A. Power-law distributions in binned empirical data. *Ann Appl Stat* 2014;8:89–119. <https://doi.org/10.1214/13-AOS710>.
- [65] Zhang Y, Wang S, Phillips P, Ji G. Binary PSO with mutation operator for feature selection using decision tree applied to spam detection. *Knowl Base Syst* 2014;64: 22–31. <https://doi.org/10.1016/j.knsys.2014.03.015>.
- [66] Alabdrabalnabi A, Gautam R, Mani Sarathy S. Machine learning to predict biochar and bio-oil yields from co-pyrolysis of biomass and plastics. *Fuel* 2022;328. <https://doi.org/10.1016/j.fuel.2022.125303>.
- [67] Leng L, Yang L, Leng S, Zhang W, Zhou Y, Peng H, et al. A review on nitrogen transformation in hydrochar during hydrothermal carbonization of biomass containing nitrogen. *Sci Total Environ* 2021;756. <https://doi.org/10.1016/j.scitotenv.2020.143679>.
- [68] Kim D, Lee K, Park KY. Upgrading the characteristics of biochar from cellulose, lignin, and xylan for solid biofuel production from biomass by hydrothermal carbonization. *J Ind Eng Chem* 2016;42:95–100. <https://doi.org/10.1016/j.jiec.2016.07.037>.
- [69] Liu T, Jiao HT, Yang L, Zhang W, Hu Y, Guo Y, et al. Co-hydrothermal carbonization of cellulose, hemicellulose, and protein with aqueous phase recirculation: insight into the reaction mechanisms on hydrochar formation. *Energy* 2022;251. <https://doi.org/10.1016/j.energy.2022.123965>.
- [70] Guedes RE, Luna AS, Torres AR. Operating parameters for bio-oil production in biomass pyrolysis: a review. *J Anal Appl Pyrolysis* 2018;129:134–49. <https://doi.org/10.1016/j.jaap.2017.11.019>.
- [71] Sekar M, Mathimani T, Alagumalai A, Chi NTL, Duc PA, Bhatia SK, et al. A review on the pyrolysis of algal biomass for biochar and bio-oil – bottlenecks and scope. *Fuel* 2021;283. <https://doi.org/10.1016/j.fuel.2020.119190>.
- [72] Isahak WNRW, Hisham MWM, Yarmo MA, Yun Hin TY. A review on bio-oil production from biomass by using pyrolysis method. *Renew Sustain Energy Rev* 2012;16:5910–23. <https://doi.org/10.1016/j.rser.2012.05.039>.
- [73] Kan T, Strezov V, Evans TJ. Lignocellulosic biomass pyrolysis: a review of product properties and effects of pyrolysis parameters. *Renew Sustain Energy Rev* 2016;57: 1126–40. <https://doi.org/10.1016/j.rser.2015.12.185>.
- [74] Saikia R, Chutia RS, Katakri R, Pant KK. Perennial grass (arundo donax L.) as a feedstock for thermo-chemical conversion to energy and materials. *Bioresour Technol* 2015;188:265–72. <https://doi.org/10.1016/j.biortech.2015.01.089>.

- [75] Vardon DR, Sharma BK, Blazina Gv, Rajagopalan K, Strathmann TJ. Thermochemical conversion of raw and defatted algal biomass via hydrothermal liquefaction and slow pyrolysis. *Bioresour Technol* 2012;109:178–87. <https://doi.org/10.1016/j.biortech.2012.01.008>.
- [76] Ruiz-Gómez N, Quispe V, Ábrego J, Atienza-Martínez M, Murillo MB, Gea G. Co-pyrolysis of sewage sludge and manure. *Waste Manag* 2017;59:211–21. <https://doi.org/10.1016/j.wasman.2016.11.013>.
- [77] Jena U, Das KC. Comparative evaluation of thermochemical liquefaction and pyrolysis for bio-oil production from microalgae. *Energy Fuel* 2011;25:5472–82. <https://doi.org/10.1021/ef201373m>.
- [78] Yang K, Wu K, Zhang H. Machine learning prediction of the yield and oxygen content of bio-oil via biomass characteristics and pyrolysis conditions. *Energy* 2022;254. <https://doi.org/10.1016/j.energy.2022.124320>.
- [79] Leng L, Huang H. An overview of the effect of pyrolysis process parameters on biochar stability. *Bioresour Technol* 2018;270:627–42. <https://doi.org/10.1016/j.biortech.2018.09.030>.
- [80] Leng L, Xu X, Wei L, Fan L, Huang H, Li J, et al. Biochar stability assessment by incubation and modelling: methods, drawbacks and recommendations. *Sci Total Environ* 2019;664:11–23. <https://doi.org/10.1016/j.scitotenv.2019.01.298>.
- [81] Leng L, Zhang W, Liu T, Zhan H, Li J, Yang L, et al. Machine learning predicting wastewater properties of the aqueous phase derived from hydrothermal treatment of biomass. *Bioresour Technol* 2022;358. <https://doi.org/10.1016/j.biortech.2022.127348>.
- [82] Zhu LT, Tang JX, Luo ZH. Machine learning to assist filtered two-fluid model development for dense gas–particle flows. *AIChE J* 2020;66. <https://doi.org/10.1002/aic.16973>.